

Regularization for Inverse Problems in Imaging: A Unifying Perspective

fNIRS Seminar, November 15, 2018

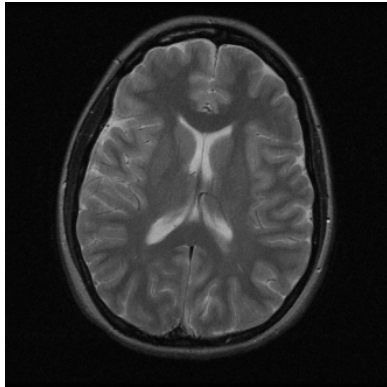
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Forward model



x : signal of interest



scan

Fourier samples

$$y_1 = \int x(\mathbf{t}) e^{-i\mathbf{k}_1^\top \mathbf{t}} d\mathbf{t} \quad (+\text{noise})$$

$$y_2 = \int x(\mathbf{t}) e^{-i\mathbf{k}_2^\top \mathbf{t}} d\mathbf{t} \quad (+\text{noise})$$

$\vdots$

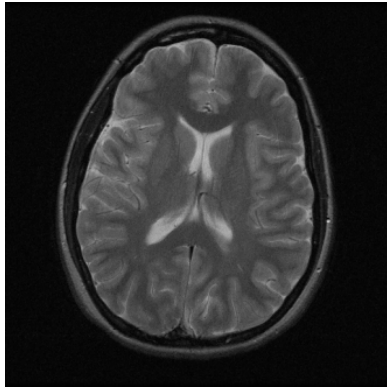
$$y_m = \int x(\mathbf{t}) e^{-i\mathbf{k}_m^\top \mathbf{t}} d\mathbf{t} \quad (+\text{noise})$$

Forward model

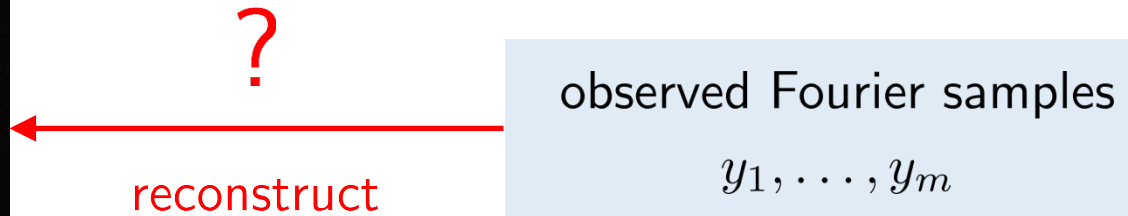
$$y_j = f_j(x) \quad (+\text{noise}), \quad j = 1, \dots, m$$

describes how observations $y_1, \dots, y_m$ are generated from signal x .

Inverse problem



x : signal of interest



Inverse problem

find x such that $f_j(x) \approx y_j$ for all $j = 1, \dots, m$

given $y_1, \dots, y_m$ (observations) and $f_1, \dots, f_m$ (forward model).

Classical inverse problem

Forward model

$$\begin{bmatrix} y \end{bmatrix} = \begin{bmatrix} A \end{bmatrix} \begin{bmatrix} x \end{bmatrix} \quad (+\text{noise}) \quad \# \text{ equations} \geq \# \text{ unknowns}$$

Q: When can be solved? How to solve?

Identifiability

A has linearly independent columns $\iff$ **unique solution**

Optimization

$$\underset{x}{\text{minimize}} \quad \|y - Ax\|_2^2$$

$\rightarrow$ **comprehensive theory** for the least squares

(guaranteed algorithms, perturbation analysis, ...)

Modern inverse problem

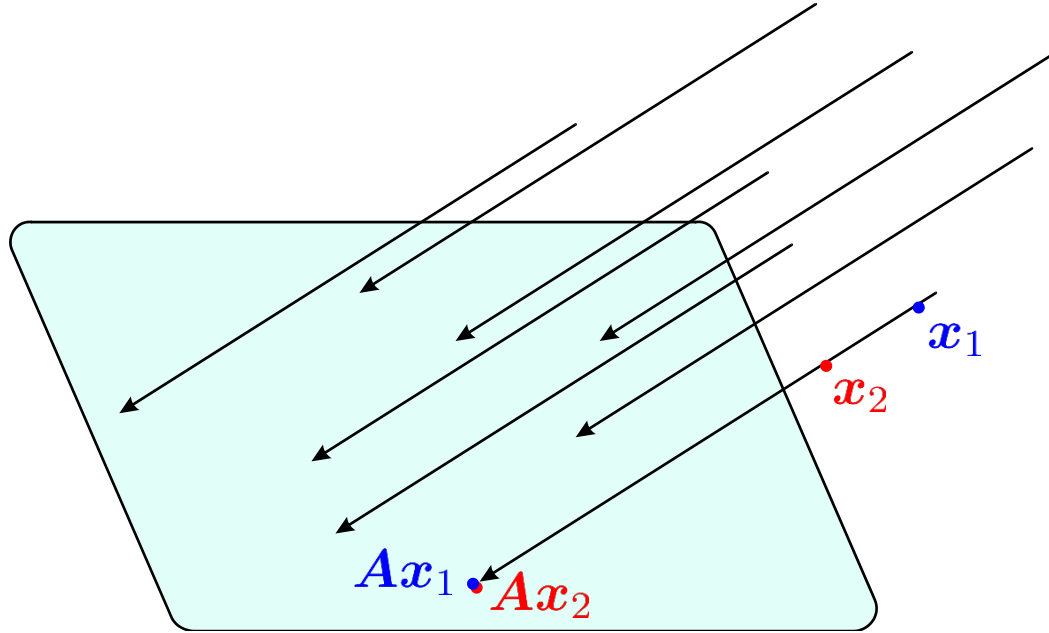
Forward model

$$\begin{bmatrix} & & & & \\ & & & & \\ & & & & \\ & & & & \\ & & & & \\ & & & & \\ & & & & \\ & & & & \\ & & & & \\ & & & & \end{bmatrix} \mathbf{A} \begin{bmatrix} & & & & \\ & & & & \\ & & & & \\ & & & & \\ & & & & \\ & & & & \\ & & & & \\ & & & & \\ & & & & \\ & & & & \end{bmatrix} \mathbf{x} \approx \begin{bmatrix} & & & & \\ & & & & \\ & & & & \\ & & & & \\ & & & & \\ & & & & \\ & & & & \\ & & & & \\ & & & & \\ & & & & \end{bmatrix} \mathbf{y}$$

equations < # unknowns

Q: When can be solved? How to solve?

Underdetermined system is ill-posed

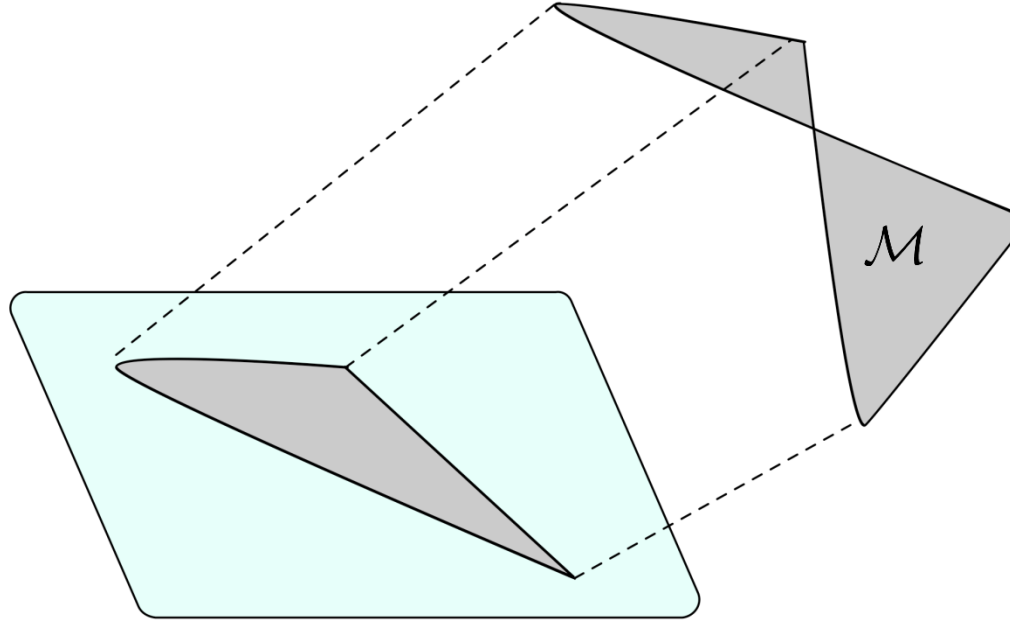


For “fat” matrix A , there are vectors x_1 and x_2 such that

$$x_1 \neq x_2 \quad \text{but} \quad Ax_1 = Ax_2.$$

$\implies$ solution **not unique**

Structured solutions



Prior model: signals of interest are structured (belong to set $\mathcal{M}$).

$\Rightarrow$ It might be possible to have

$$\text{dist}(\mathbf{A}\mathbf{x}_1, \mathbf{A}\mathbf{x}_2) \approx \text{dist}(\mathbf{x}_1, \mathbf{x}_2), \quad \forall \mathbf{x}_1, \mathbf{x}_2 \in \mathcal{M}.$$

$\Rightarrow$ **unique solution**

Q: Which $\mathcal{M}$ is interesting?

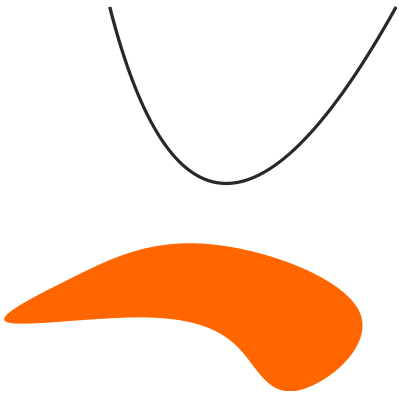
Sparsity prior



- x is s -sparse
 $\iff \|x\|_0 \leq s$ where $\|x\|_0 = \#$ nonzero entries

Sparse recovery by convex programming

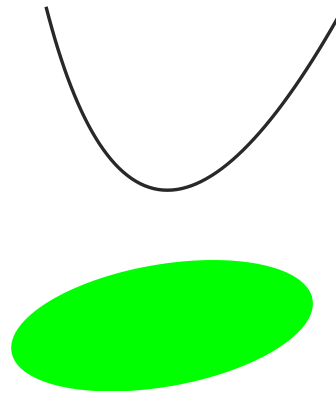
$$\begin{array}{ll}\text{minimize}_{\mathbf{x}} & \|\mathbf{y} - \mathbf{A}\mathbf{x}\|_2^2 \\ \text{subject to} & \|\mathbf{x}\|_0 \leq s\end{array}$$



nonconvex optimization

?

$$\begin{array}{ll}\text{minimize}_{\mathbf{x}} & \|\mathbf{x}\|_1 \\ \text{subject to} & \mathbf{A}\mathbf{x} = \mathbf{y}\end{array}$$



convex optimization

- comprehensive theory
- algorithms finding global minimizer(s)

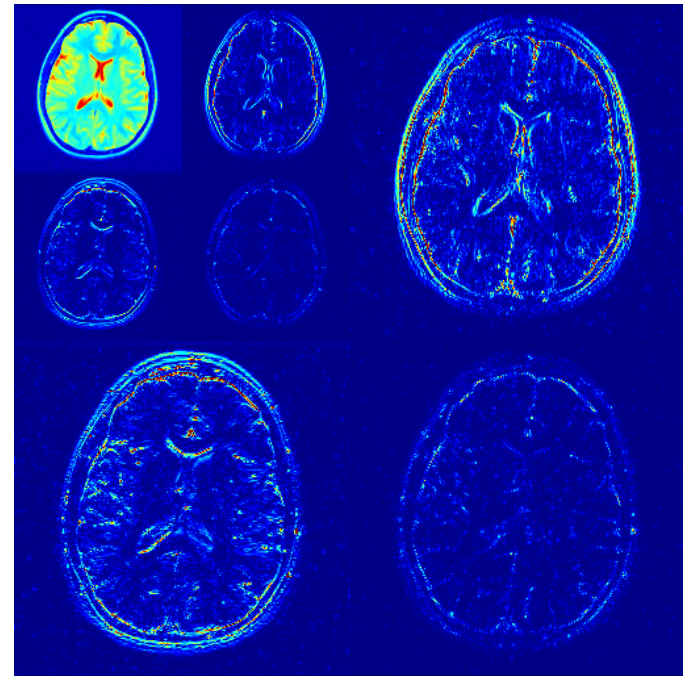
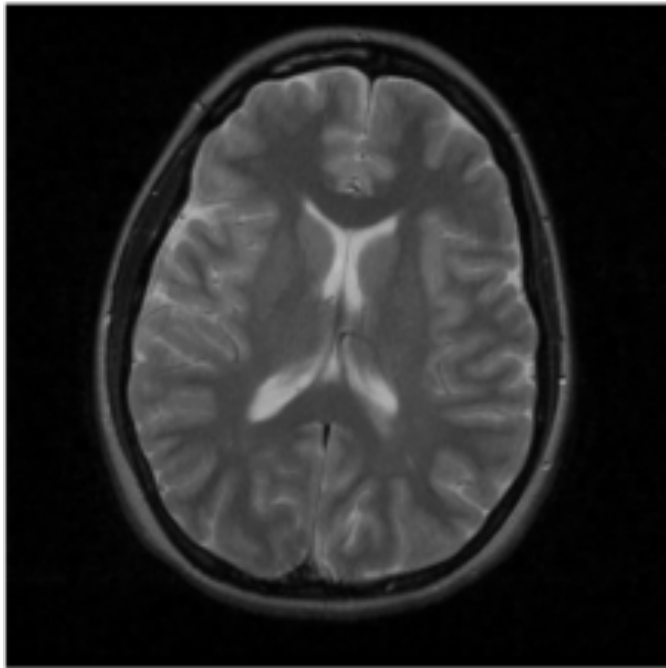
Compressed sensing theory

$$\begin{array}{ll} \underset{\mathbf{x}}{\text{minimize}} & \|\mathbf{x}\|_1 \\ \text{subject to} & \mathbf{A}\mathbf{x} = \mathbf{A}\mathbf{x}_\# \end{array}$$

- $\|\mathbf{x}_\#\|_0 \leq s$
 - $\mathbf{A}$ = randomly selected rows of DFT matrix
w.r.t. the **uniform distribution**
- $\implies \mathbf{x}_\#$ is the unique minimizer (with high probability)

Transform sparsity

In applications, sparsity is typically observed only through a certain transform.



Transform-sparse compressed sensing

signal domain

sparsity domain



- sparsifying transform $\Phi : \mathbb{C}^d \rightarrow \mathbb{C}^N$

$$\|\Phi f_{\#}\|_0 \leq s$$

signal domain

sampling domain



- sensing transform $\Psi : \mathbb{C}^d \rightarrow \mathbb{C}^n$

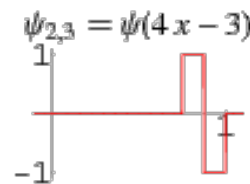
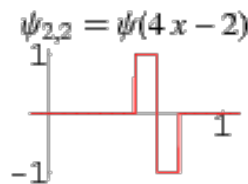
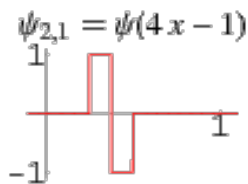
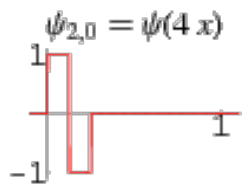
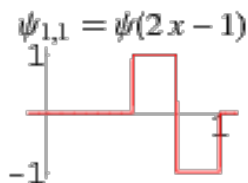
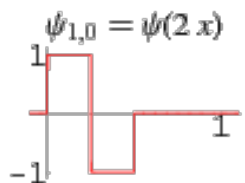
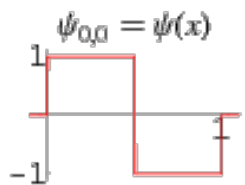
$$S_{\Omega} \Psi f_{\#}: \text{measurements of } f_{\#}$$

recovery by convex program

$$\begin{array}{ll} \underset{f \in \mathbb{C}^d}{\text{minimize}} & \|\Phi f\|_1 \\ \text{subject to} & S_{\Omega} \Psi f = S_{\Omega} \Psi f_{\#} \end{array}$$

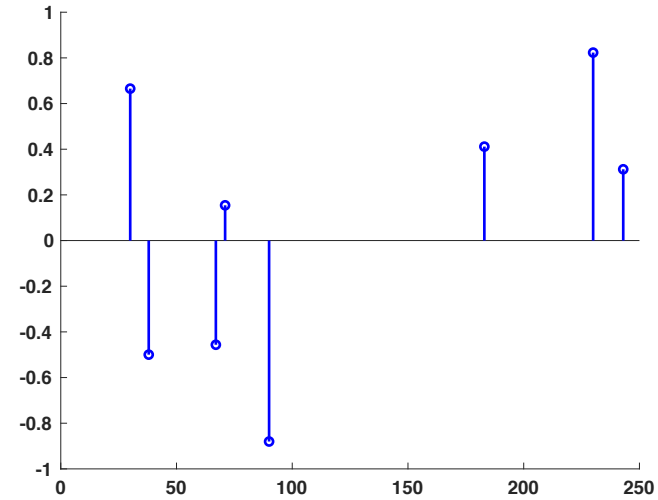
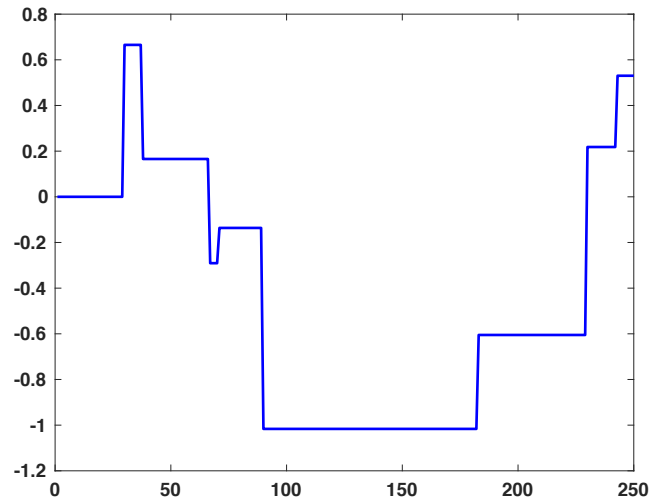
Sparsifying transform: Haar wavelet

$$H_8 = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & -1 & -1 & -1 & -1 \\ 1 & 1 & -1 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & -1 & -1 \\ 1 & -1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & -1 \end{bmatrix}.$$



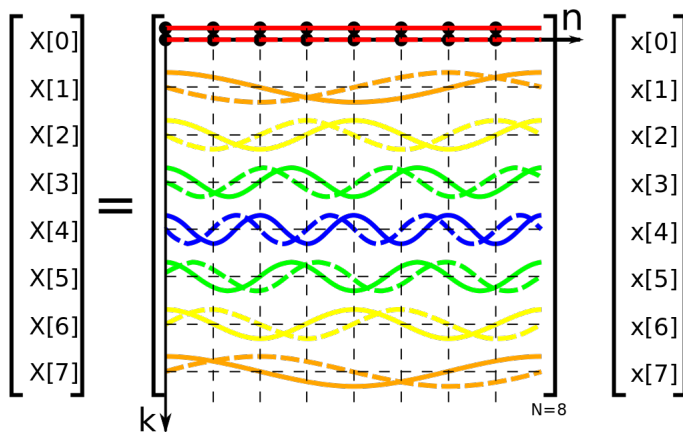
Sparsifying transform: differencing

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & -1 \\ -1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} x[0] \\ x[1] \\ x[2] \\ x[3] \\ x[4] \\ x[5] \\ x[6] \\ x[7] \end{bmatrix} = \begin{bmatrix} x[0] - x[7] \\ x[1] - x[0] \\ x[2] - x[1] \\ x[3] - x[2] \\ x[4] - x[3] \\ x[5] - x[4] \\ x[6] - x[5] \\ x[7] - x[6] \end{bmatrix}$$



Sensing transform: Discrete Fourier Transform

$$W = \begin{bmatrix} \omega^0 & \omega^0 & \omega^0 & \omega^0 & \omega^0 & \omega^0 & \omega^0 & \omega^0 \\ \omega^0 & \omega^1 & \omega^2 & \omega^3 & \omega^4 & \omega^5 & \omega^6 & \omega^7 \\ \omega^0 & \omega^2 & \omega^4 & \omega^6 & \omega^8 & \omega^{10} & \omega^{12} & \omega^{14} \\ \omega^0 & \omega^3 & \omega^6 & \omega^9 & \omega^{12} & \omega^{15} & \omega^{18} & \omega^{21} \\ \omega^0 & \omega^4 & \omega^8 & \omega^{12} & \omega^{16} & \omega^{20} & \omega^{24} & \omega^{28} \\ \omega^0 & \omega^5 & \omega^{10} & \omega^{15} & \omega^{20} & \omega^{25} & \omega^{30} & \omega^{35} \\ \omega^0 & \omega^6 & \omega^{12} & \omega^{18} & \omega^{24} & \omega^{30} & \omega^{36} & \omega^{42} \\ \omega^0 & \omega^7 & \omega^{14} & \omega^{21} & \omega^{28} & \omega^{35} & \omega^{42} & \omega^{49} \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & \omega & -i & -i\omega & -1 & -\omega & i & i\omega \\ 1 & -i & -1 & i & 1 & -i & -1 & i \\ 1 & -i\omega & i & \omega & -1 & i\omega & -i & -\omega \\ 1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 \\ 1 & -\omega & -i & i\omega & -1 & \omega & i & -i\omega \\ 1 & i & -1 & -i & 1 & i & -1 & -i \\ 1 & i\omega & i & -\omega & -1 & -i\omega & -i & \omega \end{bmatrix}$$



$$\omega = e^{-j\pi/4}$$

Abstraction

- sparsifying transform $\Phi : \mathbb{C}^d \rightarrow \mathbb{C}^N$

$$\|\Phi f_{\#}\|_0 \leq s$$

- sensing transform $\Psi : \mathbb{C}^d \rightarrow \mathbb{C}^n$

$$S_{\Omega} \Psi f_{\#}: \text{measurements of } f_{\#}$$

assume Ψ is injective

- $x_{\#} = \Psi f_{\#} \implies f_{\#} = \Psi^{\dagger} x_{\#} \implies \Phi f_{\#} = \Phi \Psi^{\dagger} x_{\#}$

- $T = \Phi \Psi^{\dagger} : \mathbb{C}^n \rightarrow \mathbb{C}^N$

sampling domain

sparsity domain

recovery by convex program

$$\begin{array}{ll} \underset{f \in \mathbb{C}^d}{\text{minimize}} & \|\Phi f\|_1 \\ \text{subject to} & S_{\Omega} \Psi f = S_{\Omega} \Psi f_{\#} \end{array}$$



$$\begin{array}{ll} \underset{x \in \mathbb{C}^n}{\text{minimize}} & \|Tx\|_1 \\ \text{subject to} & S_{\Omega} x = S_{\Omega} x_{\#} \end{array}$$

Max incoherence

sample domain

sparsity domain


$$\mathbf{T} : \mathbb{C}^n \rightarrow \mathbb{C}^N$$

injective linear transform

$$\mathbf{T}^\dagger : \mathbb{C}^N \rightarrow \mathbb{C}^n$$

pseudo inverse of $\mathbf{T}$

$$\mathbf{T}^\dagger \mathbf{T} = \mathbf{I}_n$$

incoherence parameters at sample index k

$$\mu_k := n \|\mathbf{T} \mathbf{e}_k\|_\infty^2$$

$$\tilde{\mu}_k := n \|\mathbf{T}^{\dagger*} \mathbf{e}_k\|_\infty^2$$

$$\mathbf{T}^{\dagger*} : \mathbb{C}^n \rightarrow \mathbb{C}^N$$

adjoint of $\mathbf{T}^\dagger$

max incoherence

$$\mu = \max_{1 \leq k \leq n} \max(\mu_k, \tilde{\mu}_k)$$

Recovery from uniform-density random samples

- uniform-density random index ω

$$\mathbb{P}(\omega = k) = \frac{1}{n}, \quad k = 1, \dots, n$$

- $\omega_1, \dots, \omega_m$: independent copies of ω

Theorem (uniform-density random sampling)

If $m > \frac{C(1 + \beta)\mu s}{1 - \|\mathbf{T}^* \mathbf{T} - \mathbf{I}_n\|} [\log N + \log (\|\mathbf{T}\|_{1 \rightarrow 2} \|\mathbf{T}^{\dagger*}\|_{1 \rightarrow 2})]$,

then with probability $1 - e^{-\beta} - 3/n$, $\mathbf{g}_{\#}$ is the unique minimizer to

$$\underset{\mathbf{x} \in \mathbb{C}^n}{\text{minimize}} \quad \|\mathbf{T}\mathbf{x}\|_1 \quad \text{subject to} \quad \mathbf{S}_{\Omega}\mathbf{x} = \mathbf{S}_{\Omega}\mathbf{x}_{\#}.$$

Average incoherence

sample domain

sparsity domain


$$\mathbf{T} : \mathbb{C}^n \rightarrow \mathbb{C}^N$$

injective linear transform

$$\mathbf{T}^\dagger : \mathbb{C}^N \rightarrow \mathbb{C}^n$$

pseudo inverse of $\mathbf{T}$

$$\mathbf{T}^\dagger \mathbf{T} = \mathbf{I}_n$$

incoherence parameters at sample index k

$$\mu_k := n \|\mathbf{T} \mathbf{e}_k\|_\infty^2$$

$$\tilde{\mu}_k := n \|\mathbf{T}^{\dagger*} \mathbf{e}_k\|_\infty^2$$

$$\mathbf{T}^{\dagger*} : \mathbb{C}^n \rightarrow \mathbb{C}^N$$

adjoint of $\mathbf{T}^\dagger$

max incoherence

$$\mu = \max_{1 \leq k \leq n} \max(\mu_k, \tilde{\mu}_k)$$

average incoherence

$$\bar{\mu} = \frac{1}{n} \sum_{k=1}^n \sqrt{\mu_k \tilde{\mu}_k}$$

Recovery from variable-density random samples

- variable-density random index ω

$$\mathbb{P}(\omega = k) = \frac{\sqrt{\mu_k \tilde{\mu}_k}}{\sum_{j=1}^n \sqrt{\mu_j \tilde{\mu}_j}}, \quad k = 1, \dots, n$$

- $\omega_1, \dots, \omega_m$: independent copies of ω

Theorem (variable-density random sampling)

If $m > \frac{C(1 + \beta)\bar{\mu}s}{1 - \|\mathbf{T}^* \mathbf{T} - \mathbf{I}_n\|} [\log N + \log (\|\mathbf{T}\|_{1 \rightarrow 2} \|\mathbf{T}^{\dagger*}\|_{1 \rightarrow 2})]$,
then with probability $1 - e^{-\beta} - 3/n$, $\mathbf{g}_{\#}$ is the unique minimizer to

$$\underset{\mathbf{x} \in \mathbb{C}^n}{\text{minimize}} \quad \|\mathbf{T}\mathbf{x}\|_1 \quad \text{subject to} \quad \mathbf{S}_{\Omega}\mathbf{x} = \mathbf{S}_{\Omega}\mathbf{x}_{\#}.$$

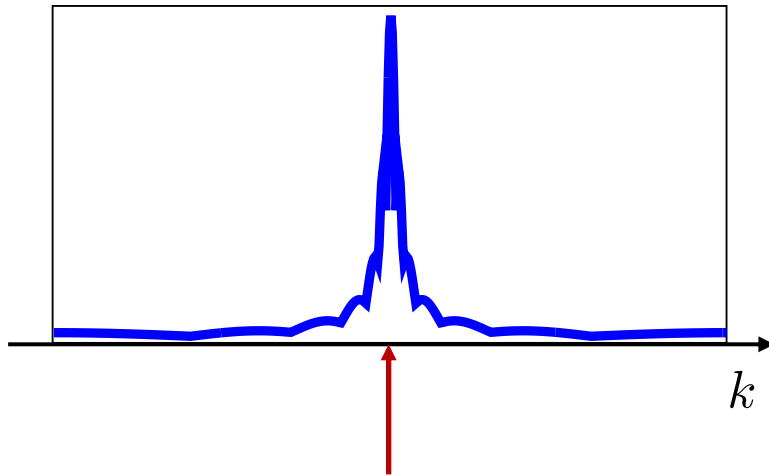
Example 1: Haar wavelet

Φ : Haar DWT, Ψ : DFT

$\sqrt{\mu_k \tilde{\mu}_k}$ is large only for a few low-frequency k s

$$\implies \bar{\mu} \ll \mu$$

corresponding variable-density



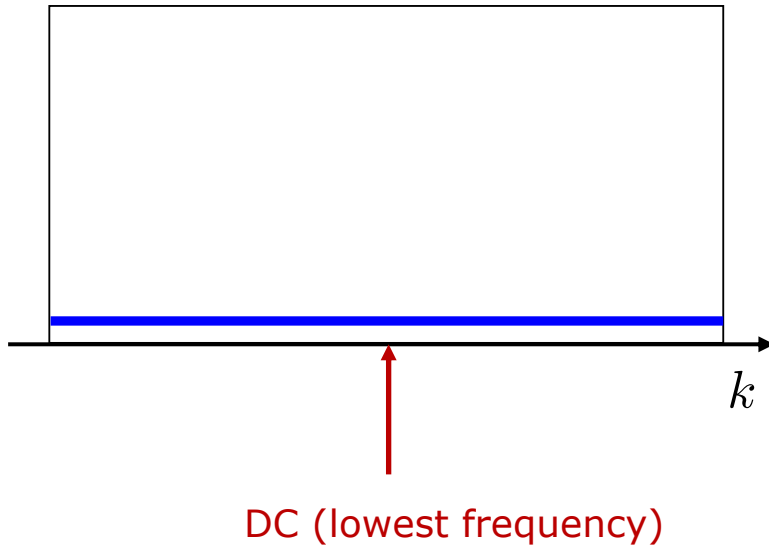
DC (lowest frequency)

Example 2: total variation (TV)

Φ : differencing (for total variation), Ψ : DFT

$\sqrt{\mu_k \tilde{\mu}_k}$ is the same for all $k \implies \bar{\mu} = \mu$

corresponding variable-density is indeed the uniform distribution



Remark on DFT + TV

Φ : sparsifying, Ψ : sensing $\longrightarrow T = \Phi \Psi^\dagger$

$$\Phi = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & -1 \\ -1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -1 & 1 \end{bmatrix} \quad \underbrace{\text{first column of } \Psi^\dagger}_{\eta} = \frac{1}{\sqrt{8}} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

differencing

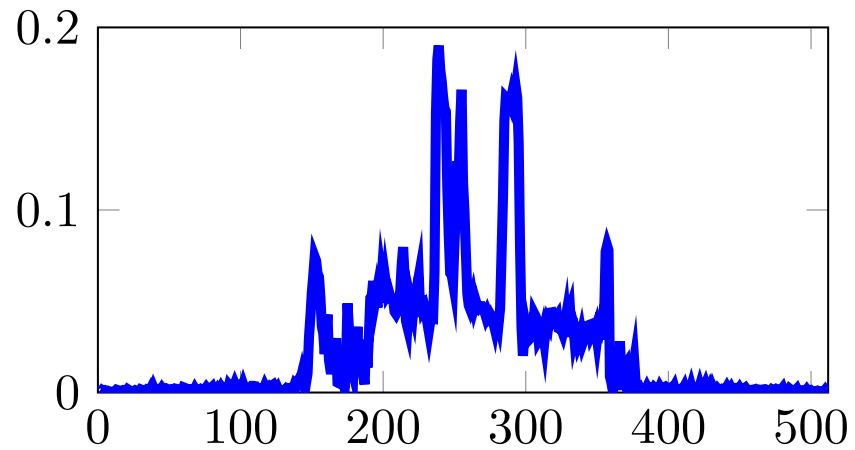
DC component

$\Phi \eta = \mathbf{0} \implies$ first column of T is zero

Therefore we always need to sample the DC component.

Design sampling density only with nonzero columns $\rightarrow$ uniform density

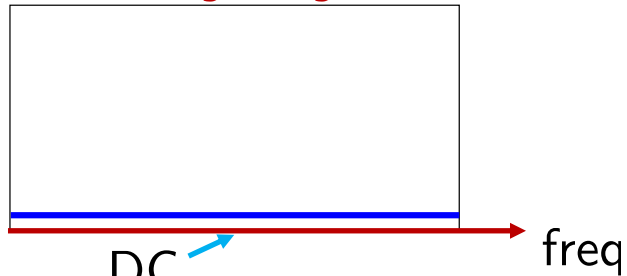
Validation on phantom signal



Here we consider the 1D case and used one horizontal line of the left image.

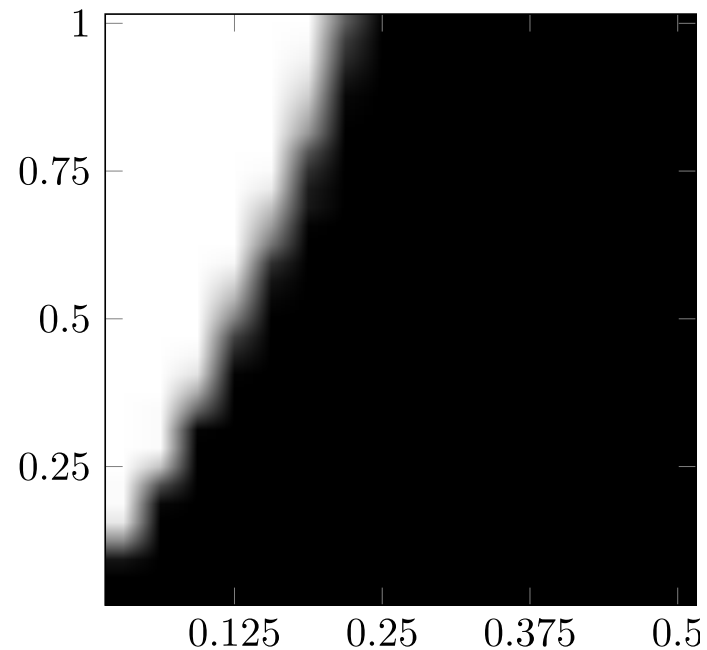
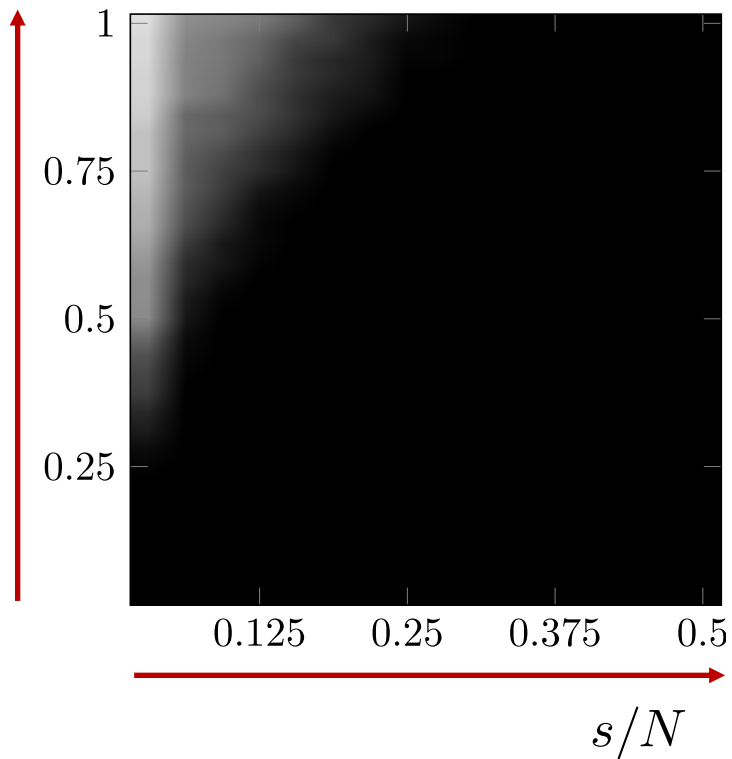
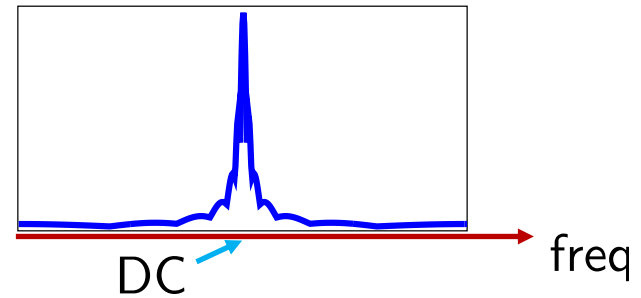
Example 1: Haar wavelet sparsity

wrong design



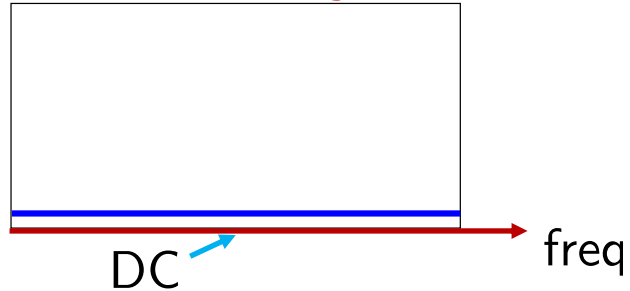
m/n

correct design



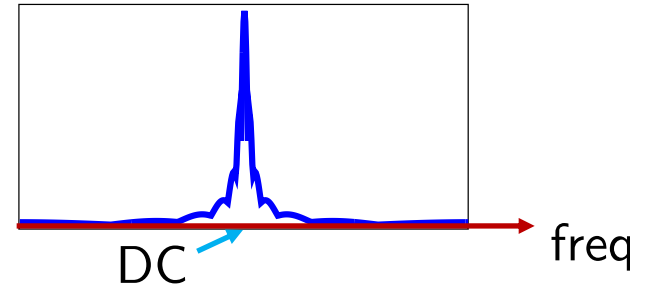
Example 2: total variation

correct design



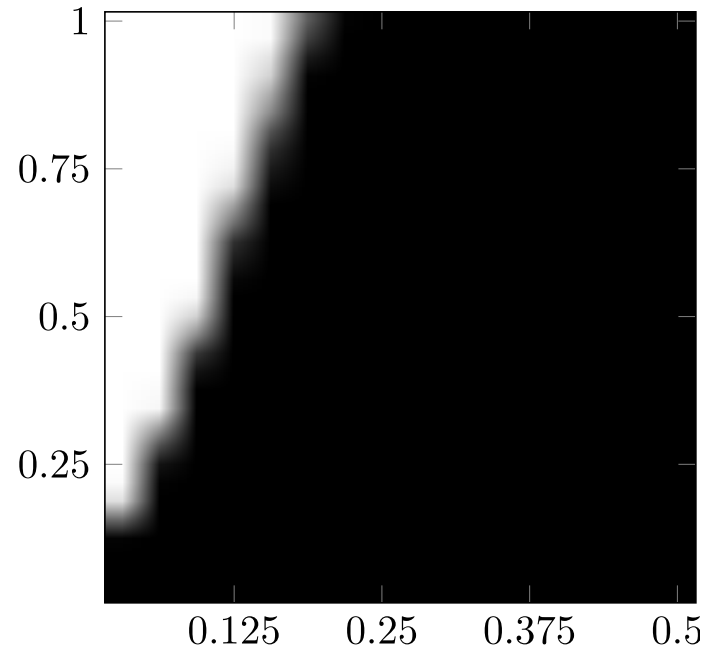
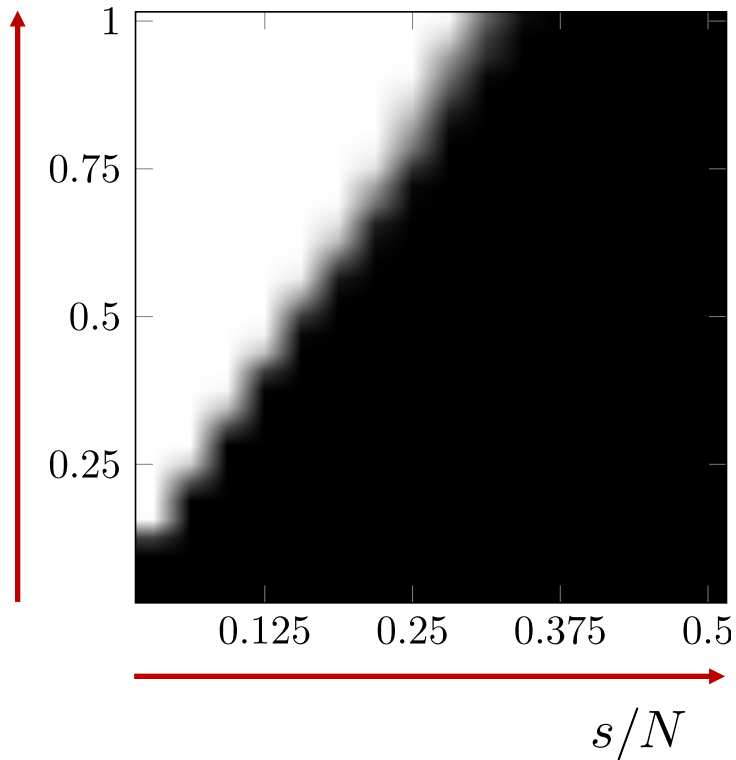
m/n

wrong design



DC

freq



Group sparsity

- $\mathcal{G} = \{\mathcal{G}_1, \mathcal{G}_2, \dots, \mathcal{G}_N\}$ is a partition of $\{1, 2, \dots, L\}$ if

$$\bigcup_{j=1}^N \mathcal{G}_j = \{1, 2, \dots, L\} \quad \text{and} \quad \mathcal{G}_j \cap \mathcal{G}_{j'} = \emptyset \quad \forall j \neq j'$$

- $\mathbf{z} \in \mathbb{C}^L$ is $(\mathcal{G}, s)$ -sparse if there is $J \subset \{1, 2, \dots, N\}$ s.t.

$$\text{supp}(\mathbf{z}) \subset \bigcup_{j \in J} \mathcal{G}_j \quad \text{and} \quad |J| \leq s$$

Group ℓ_1 norm

- coordinate projection

$$(\mathbf{\Pi}_{\mathcal{G}_j} \mathbf{z})[k] := \begin{cases} \mathbf{z}[k] & k \in \mathcal{G}_j \\ 0 & \text{else} \end{cases}$$

- $\mathcal{G}$ - ℓ_1 -norm

$$\|\mathbf{z}\|_{\mathcal{G},1} := \sum_{j=1}^N \|\mathbf{\Pi}_{\mathcal{G}_j} \mathbf{z}\|_2$$

Example: joint sparsity

- $L = 2N$
- $\mathbf{z} = \begin{bmatrix} \mathbf{z}_1 \\ \mathbf{z}_2 \end{bmatrix}, \quad \mathbf{z}_1, \mathbf{z}_2 \in \mathbb{C}^N$
- $\mathcal{G}_j = \{j, j + N\}$ for $j = 1, \dots, N$
- $|\text{supp}(\mathbf{z}_1) \cup \text{supp}(\mathbf{z}_2)| \leq s \implies \mathbf{z}$ is $(\mathcal{G}, s)$ -sparse
- $\mathcal{G}$ - ℓ_1 -norm
$$\|\mathbf{z}\|_{\mathcal{G},1} := \sum_{j=1}^N \|\mathbf{\Pi}_{\mathcal{G}_j} \mathbf{z}\|_2 = \sum_{j=1}^N \sqrt{|\mathbf{z}_1[j]|^2 + |\mathbf{z}_2[j]|^2}$$

Transform-group-sparse recovery

sparsity model

$Tx_{\#}$ is $(\mathcal{G}, s)$ -sparse

recovery by convex program

$$\underset{x \in \mathbb{C}^n}{\text{minimize}} \quad \|Tx\|_{\mathcal{G},1} \quad \text{subject to} \quad S_{\Omega}x = S_{\Omega}x_{\#}.$$

Group average incoherence

sample domain group sparsity domain


$$\mathbf{T} : \mathbb{C}^n \rightarrow \mathbb{C}^L$$

group incoherence parameters at sample index k

$$\mu_{\mathcal{G},k} := n \max_{1 \leq j \leq N} \|\mathbf{\Pi}_{\mathcal{G}_j} \mathbf{T} \mathbf{e}_k\|_2^2$$

$$\tilde{\mu}_{\mathcal{G},k} := n \max_{1 \leq j \leq N} \|\mathbf{\Pi}_{\mathcal{G}_j} \mathbf{T}^{\dagger*} \mathbf{e}_k\|_2^2$$

group average incoherence

$$\bar{\mu}_{\mathcal{G}} := \frac{1}{n} \sum_{k=1}^n \sqrt{\mu_{\mathcal{G},k} \tilde{\mu}_{\mathcal{G},k}}$$

Recovery from variable-density random samples

- variable-density random index ω

$$\mathbb{P}(\omega = k) = \frac{\sqrt{\mu_{\mathcal{G},k} \tilde{\mu}_{\mathcal{G},k}}}{\sum_{j=1}^n \sqrt{\mu_{\mathcal{G},j} \tilde{\mu}_{\mathcal{G},j}}}, \quad k = 1, \dots, n$$

- $\omega_1, \dots, \omega_m$: independent copies of ω

Theorem (variable-density random sampling)

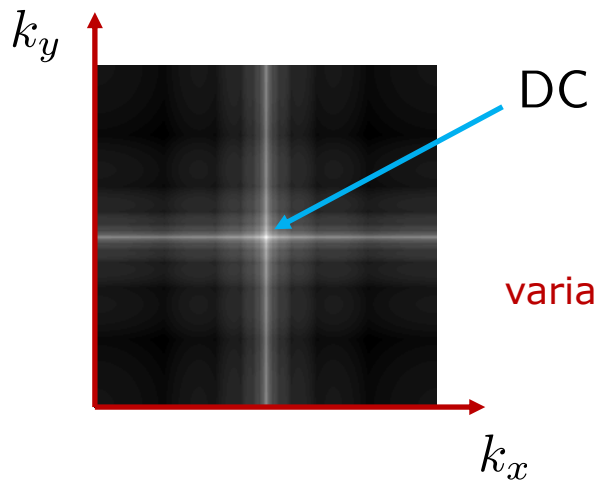
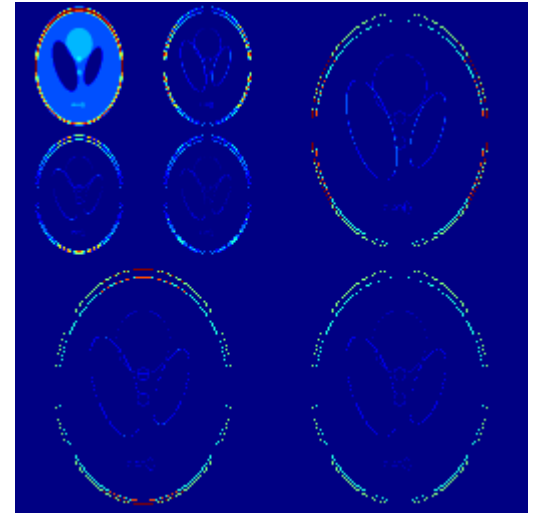
If $m > \frac{C(1 + \beta)\bar{\mu}_{\mathcal{G}}s}{1 - \|\mathbf{T}^*\mathbf{T} - \mathbf{I}_n\|} [\log N + \log (\|\mathbf{T}\|_{1 \rightarrow 2} \|\mathbf{T}^{\dagger*}\|_{1 \rightarrow 2})]$,
then with probability $1 - e^{-\beta} - 3/n$, $\mathbf{g}_{\#}$ is the unique minimizer to

$$\underset{\mathbf{x} \in \mathbb{C}^n}{\text{minimize}} \quad \|\mathbf{T}\mathbf{x}\|_{\mathcal{G},1} \quad \text{subject to} \quad \mathbf{S}_{\Omega}\mathbf{x} = \mathbf{S}_{\Omega}\mathbf{x}_{\#}.$$

2D transform sparsity: wavelet



$$\Phi = 2D \text{ DWT}$$



variable density designed by incoherence with 2D DFT

2D transform sparsity: total variation

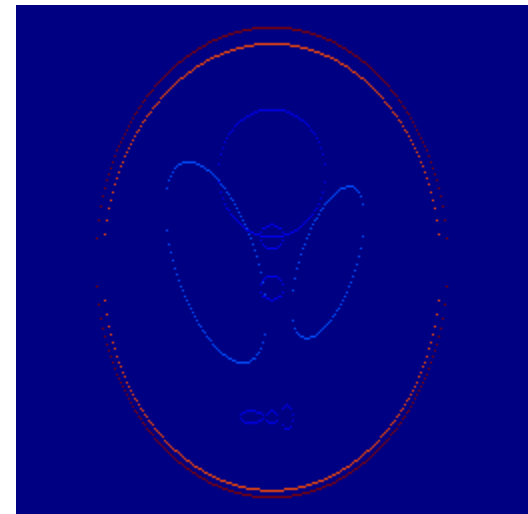
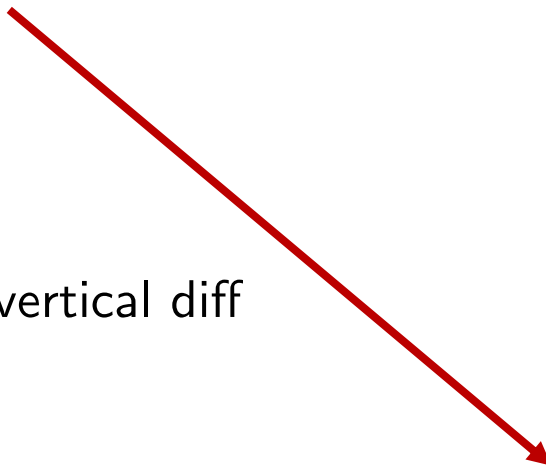


$\Phi_1 = \text{horizontal diff}$



$$\Phi = \begin{bmatrix} \Phi_1 \\ \Phi_2 \end{bmatrix}$$

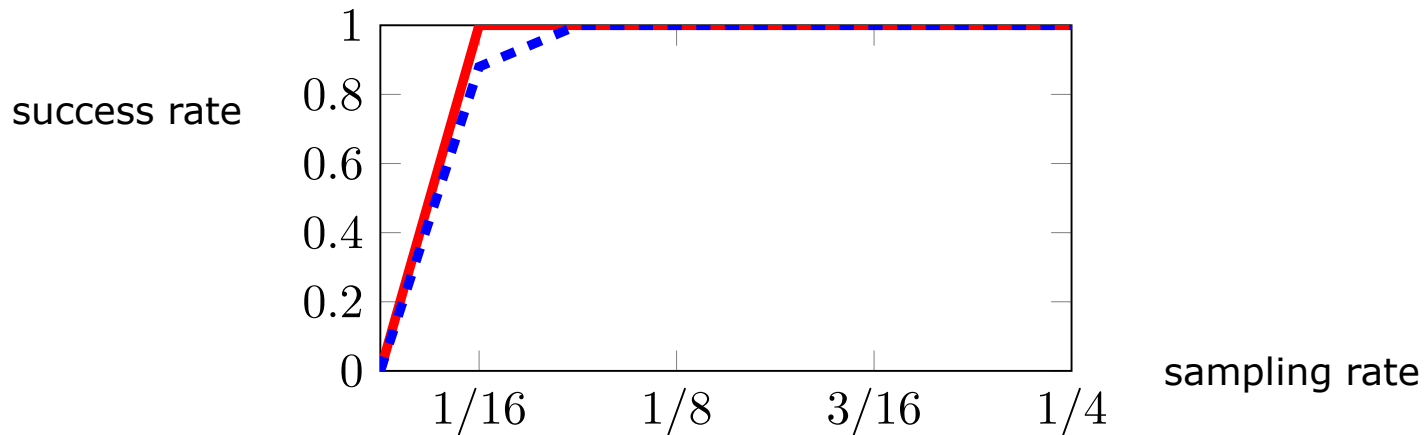
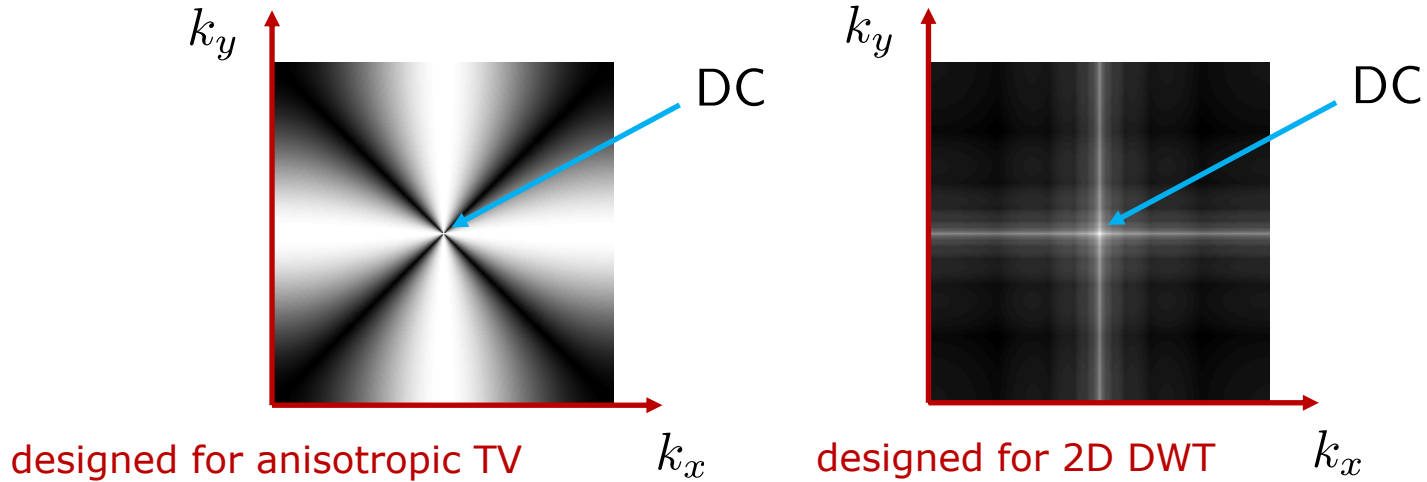
$\Phi_2 = \text{vertical diff}$



- $\|\Phi x\|_1 = \text{anisotropic total variation}$
- $\|\Phi x\|_{\mathcal{G},1} = \text{isotropic total variation}$

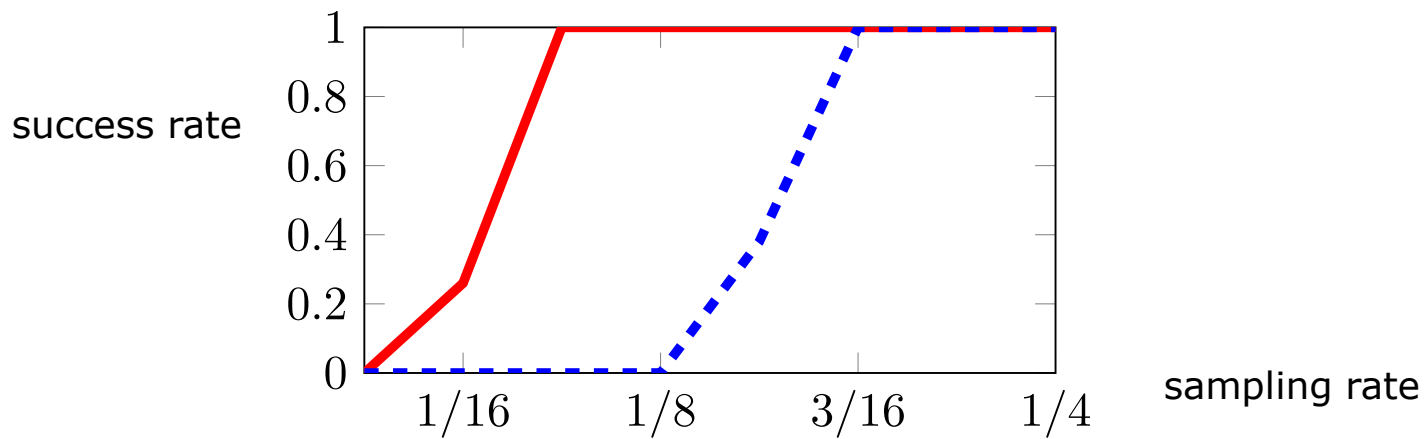
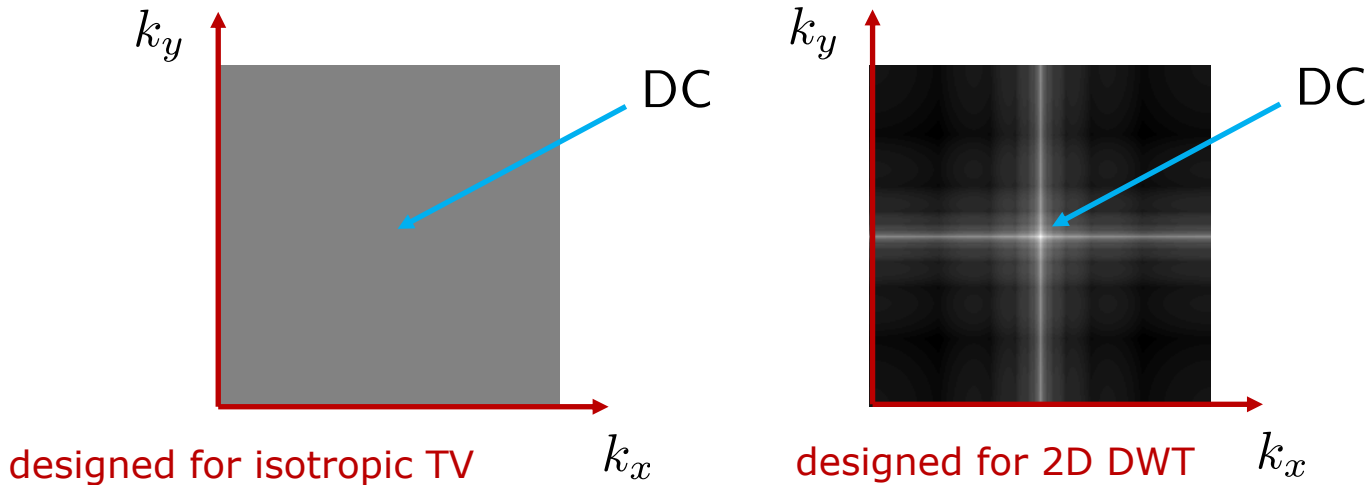
joint sparsity

Example 3: Anisotropic TV minimization



performance with **correct design** vs performance with **mismatched design**

Example 4: Isotropic TV minimization



performance with **correct design** vs performance with **mismatched design**

Analogous results for low-rankness

$$\mathbf{T} : \mathbb{C}^n \rightarrow \mathbb{C}^{d_1 \times d_2}$$

- assumption: $\mathbf{T}\mathbf{x}_\#$ is a low-rank matrix

$$\underset{\mathbf{x} \in \mathbb{C}^n}{\text{minimize}} \quad \|\mathbf{T}\mathbf{x}\|_* \quad \text{subject to} \quad \mathbf{S}_\Omega \mathbf{x} = \mathbf{S}_\Omega \mathbf{x}_\#.$$

$\|\mathbf{T}\mathbf{x}\|_*$ = sum of singular values of $\mathbf{T}\mathbf{x}$ a.k.a. nuclear norm

relevant applications: off-the-grid sparsity recovery (superresolution)

Fourier samples are mapped to a low-rank Hankel matrix

Conclusion

- theory for designing sampling density for transform sparse recovery
- theory is consistent with empirical performance
- relevant projects in progress:
 - calibration-free dynamic pMRI
 - inverse problem with piecewise polynomial models

